

**Started on** Friday, 21 August 2026, 4:39 PM**State** Finished**Completed on** Friday, 21 August 2026, 4:39 PM**Time taken** 10 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeros(int arr[], int left, int right)
4  {
5      if (left == right)
6      {
7          if (arr[left] == 0)
8          {
9              return 1;
10         }
11         return 0;
12     }
13
14     int mid = left + (right - left) / 2;
15
16     int leftCount = countZeros(arr, left, mid);
17     int rightCount = countZeros(arr, mid + 1, right);
18
19     return leftCount + rightCount;
20 }
21
22 int main()
23 {
24     int n;
25     if (scanf("%d", &n) != 1 || n <= 0) return 0;
26
27     int arr[n];
28     for (int i = 0; i < n; i++)
29     {
30         if (scanf("%d", &arr[i]) != 1) return 0;
31     }
32
33     int totalZeros = countZeros(arr, 0, n - 1);
34
35     printf("%d\n", totalZeros);
36     return 0;
37 }
38

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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